

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of BSc Physic Examination March 2018

PD 202 Thermal Physics

Date: 27-03-2018 Day: Tuesday Time: 01.30PM To 02:00PM Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. 20

1. One container is filled with argon gas and another with helium gas. Both containers are at the same temperature. Which atoms have the higher rms speed?
(a) argon
(b) helium
(c) they have the same speed
(d) not stable
2. If you are asked to make a very sensitive glass thermometer, which of the following working fluids would you choose?
(a) mercury
(b) alcohol
(c) gasoline
(d) glycerin
3. Two spheres are made of the same metal and have the same radius, but one is hollow and the other is solid. The spheres are taken through the same temperature increase. Which sphere expands more?
(a) solid sphere
(b) hollow sphere
(c) they expand by the same amount
(d) both will not expand

4. On a very cold day in upstate New York, the temperature is 225°C , which is equivalent to what temperature in Fahrenheit?
 - (a) 246°F
 - (b) 213°F
 - (c) 18°F
 - (d) 298 K
5. A container holds 0.50 m^3 of oxygen at an absolute pressure of 4.0 atm . A valve is opened, allowing the gas to drive a piston, increasing the volume of the gas until the pressure drops to 1.0 atm . If the temperature remains constant, what new volume does the gas occupy?
 - (a) 1.0 m^3
 - (b) 1.5 m^3
 - (c) 2.0 m^3
 - (d) 0.12 m^3
6. One way to cool a gas is to let it expand. When a certain gas under a pressure of $5 \times 10^6\text{ Pa}$ at 25°C is allowed to expand to 3 times its original volume, its final pressure is $1.07 \times 10^6\text{ Pa}$. What is its final temperature?
 - (a) 177°C
 - (b) 233 K
 - (c) 212 K
 - (d) 191 K
7. Suppose for a brief moment the gas molecules hitting a wall stuck to the wall instead of bouncing off the wall. How would the pressure on the wall be affected during that brief time?
 - (a) The pressure would be zero
 - (b) The pressure would be halved
 - (c) The pressure would remain unchanged
 - (d) The pressure would double
8. A rubber balloon is filled with 1 L of air at 1 atm and 300 K and is then cooled to 100 K . The rubber remains flexible as it cools. What happens to the volume of the balloon?
 - (a) It decreases to $1/6\text{ L}$
 - (b) It decreases to $1/3\text{ L}$
 - (c) It decreases to $L/\sqrt{3}$
 - (d) It is constant
9. The equipartition theorem states that, at equilibrium, each degree of freedom contributes ----- of energy per molecule.
 - (a) $3/2 k_{\text{B}}T$
 - (b) $1/2 k_{\text{B}}T$
 - (c) $k_{\text{B}}T$
 - (d) $2/3 kT$
10. The molar specific heat of a gas is measured at constant volume and found to be $11R/2$. The gas is most likely to be:
 - (a) monatomic
 - (b) diatomic
 - (c) polyatomic
 - (d) none of the above
11. An ideal gas is compressed to half its initial volume by means of several possible processes. Which of the following processes results in the most work done on the gas?
 - (a) isothermal
 - (b) adiabatic
 - (c) isobaric
 - (d) The work done is independent of the process.

12. In a reversible adiabatic process, the change of entropy
 - (a) increases
 - (b) decreases
 - (c) remains unchanged
 - (d) none of these
13. An engine does 15 kJ of work while rejecting 37 kJ to the cold reservoir. What is the efficiency of the engine?
 - (a) 0.15
 - (b) 0.29
 - (c) 0.33
 - (d) 0.45
14. Net entropy change of a system in Carnot's cycle is
 - (a) zero
 - (b) positive
 - (c) negative
 - (d) more than 1
15. A diatomic ideal gas expands adiabatically from a volume of 1 m^3 to a final volume of 3.50 m^3 . If the initial pressure is $1 \times 10^5\text{ Pa}$, what is the final pressure?
 - (a) $6.62 \times 10^4\text{ Pa}$
 - (b) $1.24 \times 10^5\text{ Pa}$
 - (c) $3.54 \times 10^4\text{ Pa}$
 - (d) $1.73 \times 10^4\text{ Pa}$
16. The change in entropy of a mole of an ideal gas, when the gas undergoes free expansion is
 - (a) positive
 - (b) negative
 - (c) zero
 - (d) none of these
17. What is the name of the following statement: "When two systems are in thermal equilibrium with a third system, then they are in thermal equilibrium with each other"?
 - (a) First Law of Thermodynamics
 - (b) Second Law of Thermodynamics
 - (c) Mechanical equivalent of heat
 - (d) Zeroth Law of Thermodynamics
18. An ideal gas is maintained at a constant pressure of 70 kPa during an isobaric process while its volume decreases by 0.20 m^3 . What is the work done by the system on its environment?
 - (a) 14 kJ
 - (b) 35 kJ
 - (c) -14 kJ
 - (d) -35 kJ
19. Entropy of fixed amount of an ideal gas:
 - (a) Decreases in every isothermal compression
 - (b) Increases in every isothermal compression
 - (c) Remains the same in every isothermal compression
 - (d) May increase or decrease in every isothermal compression
20. If an ideal gas is compressed isothermally, which of the following statements is true?
 - (a) Energy is transferred to the gas by heat
 - (b) No work is done on the gas
 - (c) The temperature of the gas increases
 - (d) The internal energy of the gas remains constant

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II Semester of BSc Physic Examination March 2018

PD 202 Thermal Physics

Date: 27-03-2018 Day: Tuesday Time: 02.00PM To 04:30PM Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION – I

Q – II Answer the following questions as directed **20**

1. Write four assumptions of kinetic theory of gases. 2
2. (i) If a glass is turned upside down and then submerged in water, what happens to the volume of the trapped air as the glass is pushed deeper under water? 2
(ii) Assuming constant temperature, does a helium balloon expand, contract, or remain at constant volume as it rises through the air?
3. (i) What happens to the tension of wires in a piano when the temperature decreases? 2
(ii) Why do vapor bubbles in a pot of boiling water get larger as they approach the surface?
4. Assuming all have the same initial volume, rank the following substances by the amount of volume expansion due to an increase in temperature, from least to most: glass, mercury, aluminum, ethyl alcohol. 2
5. Metal lids on glass jars can often be loosened by running hot water over them. Why does that work? 2
6. (i) Suppose you have 1 kg each of iron, glass, and water, and all three samples are at 10°C. 2
(a) Rank the samples from lowest to highest temperature after 100 J of energy is added to each by heat.
(ii) What effect would doubling the initial amount of liquid water have on the amount of ice melted?
7. In usually warm climates that experience an occasional hard freeze, fruit growers will spray the fruit trees with water, hoping that a layer of ice will form on the fruit. Why would such a layer be advantageous? 2
8. (i) A tile floor may feel uncomfortably cold to your bare feet, but a carpeted floor in an adjoining room at the same temperature feels warm. Why? 2
(ii) Concrete has a higher specific heat than does soil. Use this fact to explain (partially) why a city has a higher average temperature than the surrounding countryside. Would you expect evening breezes to blow from city to country or from country to city? Explain.
9. Describe Andrew's experiment on CO₂ and discuss the results obtained. 2
10. When a sealed Thermos bottle full of hot coffee is shaken, what changes, if any, take place in (a) the temperature of the coffee and (b) its internal energy? 2

SECTION – II

Q-III Answer the following questions as directed **30**

1. (i) True or False: At the same temperature, 1 mole of helium gas has the same internal energy as 1 mole of argon gas. 3
(ii) True or False: Finding the relationship between two temperature scales using knowledge of the freezing and boiling point of water in each system is equivalent to finding the equation of a straight line.

- (iii) True or False: When an ideal gas expands at constant pressure, the change in the internal energy must be positive.
2. What are some factors that affect the efficiency of automobile engines? 3
 3. An ideal gas absorbs 5×10^3 J of energy while doing 2×10^3 J of work on the environment during a constant pressure process. (a) Compute the change in the internal energy of the gas. (b) If the internal energy now drops by 4.5×10^3 J and 7.5×10^3 J is expelled from the system, find the change in volume, assuming a constant pressure process at 1.01×10^5 Pa. 3
 4. A steam engine has a boiler that operates at 5×10^2 K. The energy from the boiler changes water to steam, which drives the piston. The temperature of the exhaust is that of the outside air, 3×10^2 K. (a) What is the engine's efficiency if it's an ideal engine? (b) If the 3.5×10^3 J of energy is supplied from the boiler, find the energy transferred to the cold reservoir and the work done by the engine on its environment. 3
 5. A 1.0-mol sample of an ideal gas is kept at 0.0°C during an expansion from 3.0 L to 10.0 L. 3
 - (i) How much work is done on the gas during the expansion?
 - (ii) How much energy transfer by heat occurs with the surroundings in this process?
 - (iii) If the gas is returned to the original volume by means of an isobaric process, how much work is done on the gas?

OR

- A block of ice at 273 K is put in thermal contact with a container of steam at 373 K, converting 25 g of ice to water at 273 K while condensing some of the steam to water at 373 K. (a) Find the change in entropy of the ice. (b) Find the change in entropy of the steam. (c) Find the change in entropy of the Universe. Use: $L_f = 3.33 \times 10^5$ J
6. A cylinder contains 2 mol of helium gas at 20°C . Assume the helium behaves like an ideal gas. (a) Find the total internal energy of the system. (b) What is the average kinetic energy per molecule? The molar mass of helium is equal to 4×10^{-3} kg/mol. 3

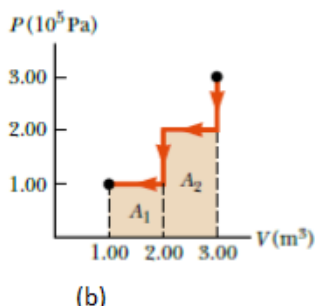
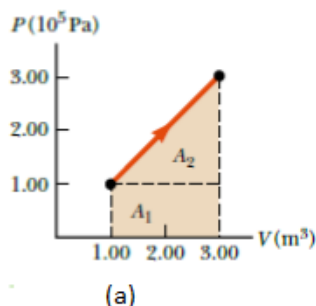
OR

- A current of 10A is passed through the resistor 100Ω resistance for 300 S. If the resistor is held at a constant temperature of 300 K.
- (i) What is the change in the entropy of the resistor?
 - (ii) What is the change in entropy of the universe?
 - (iii) What is the change in the Helmholtz free energy of the universe?
7. An ideal gas at 20°C and a pressure of 1.5×10^5 Pa is in a container having a volume of 1 L. (a) Determine the number of moles of gas in the container. (b) The gas pushes against a piston, expanding to twice its original volume, while the pressure falls to atmospheric pressure. Find the final temperature. 4
 8. (i) A steel railroad track has a length of 30 m when the temperature is 0°C . What is its length on a hot day when the temperature is 40°C ? 4
 (ii) Suppose the track is nailed down so that it can't expand. What stress results in the track due to the temperature change?

OR

Find the numeric value of the work done on the gas in:

- (i) Figure (a) and (ii) Figure (b)



9. A monatomic ideal gas has a temperature $T = 3 \times 10^2$ K and a constant volume of 1.50 L. If there are 5 moles of gas, **(a)** how much thermal energy must be added in order to raise the temperature of the gas to 3.8×10^2 K? **(b)** Calculate the change in pressure of the gas, ΔP . **(c)** How much thermal energy would be required if the gas were ideal and diatomic? **(d)** Calculate the change in the pressure for the diatomic gas. 4

Use the constants:

$$N_A = 6.02 \times 10^{23} / \text{mol}$$

$$k_B = 1.38 \times 10^{-23} \text{ J/K}$$

$$R = 8.31 \text{ J/mol.K}$$

$$\text{Specific heat capacity of water} = 4190 \text{ J/kg.K,}$$